

09

Scientists dissected gills from several species of fish. They recorded:

- the mass of the whole fish
- the total number of gill filaments
- the mean length of one filament
- the mean number of lamellae per mm
- the mean surface area of one lamella.

09.1

It was not possible for the scientists to measure the length of every filament and the surface area of every lamella.

Suggest how they collected data to give a reliable mean for these variables.

[2 marks]

09.2

From these measurements, the scientists calculated the total surface area of the gas exchange surface on the gills of each fish species.

Calculate the total surface area of the gills of a fish with the following measurements:

- total number of gill filaments = 595
- mean length of one filament = 2.86 mm
- mean number of lamellae per mm = 16
- mean surface area of one lamella = 0.66 mm²

Give your answer in mm² **and** to an appropriate number of significant figures.

Show your working.

[2 marks]

_____ mm²

Question 9 continues on the next page

Turn over ►



09.3

Table 5 shows the scientists' data for two species of fish.

Table 5

Fish species	Mean fish mass / g	Mean total surface area of the gills / mm ²
<i>Opsanus tau</i>	305	46 100
<i>Trachurus trachurus</i>	250	252 500

One of these fish spends most of its time not moving, waiting to catch passing prey. The other species is very active, hunting mobile prey.

Suggest which of the species in **Table 5** is the very active fish species.

Explain your answer.

[2 marks]

Very active fish species _____

Explanation _____

09.4

Complete **Table 6** to show the phylogenetic classification for these two species.

[2 marks]

Table 6

Taxon	<i>Opsanus tau</i>	<i>Trachurus trachurus</i>
		Animalia
	Chordata	
Class	Actinopterygii	Actinopterygii
	Batrachoidiformes	Carangiformes
Family	Batrachoididae	Carangidae
Genus		
Species	<i>tau</i>	<i>trachurus</i>

